

Liam Huntsucker

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Professional Summary: Mechanical Engineering graduate with experience in electromechanical design, embedded systems, and RF hardware development. Skilled in developing multidisciplinary products from concept through prototyping, testing, and validation.

Technical Skills

- **Programming:** C/C++, Python, MATLAB
- **Embedded:** Raspberry Pi, Arduino, SAMD21, PlatformIO, CAN Bus, MAVLink, I²C, UART, PWM
- **CAD & Design:** Solidworks, AutoCAD, KiCad, Blender
- **Simulation & Analysis:** CST Studio Suite, VSPAero, Adams, Abaqus
- **Manufacturing & Prototyping:** 3D Printing, CNC, Soldering, Fiberglass Lay-ups, Sheet Metal

Engineering Experience

Undergraduate Research Assistant – Electromechanical & RF Systems Development

Missouri Institute for Defense & Energy, UMKC | Kansas City, MO | May 2025 - Present

UAV Gimbal Controller: C++, PlatformIO, Arduino, MAVLink, CAN Bus, SimpleFOC

- Developed an embedded motor control system for a single-axis UAV camera gimbal using field-oriented control (FOC).
- Designed and prototyped a custom electromechanical housing to integrate a single-axis camera gimbal into a Class II UAV nose cone, balancing structural support, packaging constraints, and serviceability.
- Integrated IMU sensors, encoder feedback, and autopilot telemetry to achieve closed-loop camera stabilization.
- Implemented MAVLink and CAN communication for real-time interaction with the flight controller.

24 GHz Microwave Antenna Array Design: CST Studio Suite, MATLAB,

- Designed and optimized 24 GHz microstrip patch antenna arrays using electromagnetic simulation.
- Analyzed impedance matching, radiation patterns, beamwidth, and gain performance.
- Developed RF system analyses including impedance matching, and transmit/receive gain considerations.
- Optimized antenna geometry, substrate selection, and array spacing through parametric simulation.

Additional Projects

Senior Design Hydroponics Project: Blender, Solidworks, 3D Printing, Soldering

- Designed a modular vertical hydroponic system emphasizing manufacturability, structural design, and water distribution.
- Developed CAD models, prototypes, and fabrication methods for system integration.

UAV AI Vision Tracking System: Python, MAVLink, Raspberry Pi, YOLO, OpenCV

- Developed a real-time object detection and tracking pipeline using Python, YOLO, and OpenCV.
- Integrated computer vision with UAV control systems through MAVLink communication.

Education

University of Missouri - Kansas City: B.S. in Mechanical Engineering, Product Design Emphasis

GPA: 3.56 | Dean's list (Fall 2024, Spring 2025-2026) | Discipline Distinction

- **Relevant Coursework:** FEA, Intro to Polymers/Soft Materials, Composite Materials, Manufacturing Methods, Mechanical Component Design, Heat Transfer